

Module 15: Ecology

Keys to Success & Study Guide

Learning Objectives

By the end of this module, you should be able to:

1. **Define** ecology and distinguish the levels of ecological organization (organism, population, community, ecosystem, biome, biosphere).
2. **Compare** exponential and logistic population growth and explain the concept of carrying capacity.
3. **Classify** species interactions (competition, predation, mutualism, commensalism, parasitism) and predict their effects on populations.
4. **Trace** the flow of energy through trophic levels and explain why energy transfer is inefficient.
5. **Describe** the major biogeochemical cycles (water, carbon, nitrogen) and how human activity disrupts them.
6. **Explain** why biodiversity matters and identify the major threats to it.

Key Terminology Checklist

Define these terms in your own words to ensure mastery.

- [] **Ecology**: The scientific study of interactions between organisms and their environment.
- [] **Population**: A group of individuals of the same species living in the same area at the same time.
- [] **Community**: All the populations of different species living and interacting in a particular area.
- [] **Ecosystem**: A community of organisms together with the abiotic (non-living) factors of their environment.

- [] **Carrying Capacity (K):** The maximum population size an environment can sustain indefinitely given available resources.
- [] **Trophic Level:** A feeding position in a food chain (producer, primary consumer, secondary consumer, etc.).
- [] **Keystone Species:** A species whose removal would cause a disproportionately large change in its community.
- [] **Biodiversity:** The variety of life at all levels — genetic, species, and ecosystem diversity.

Concept Check

1. Levels of Organization

- **Question:** What are the levels of ecological organization?
- **Key Answer:** Organism → Population → Community → Ecosystem → Biome → Biosphere. Each level includes the one below it plus emergent properties.

2. Population Growth

- **Question:** What is the difference between exponential and logistic growth?
- **Key Answer:** Exponential growth (J-curve) occurs when resources are unlimited — the population grows without bound. Logistic growth (S-curve) occurs when resources are limited — growth slows as population approaches carrying capacity (K).

3. Energy Flow

- **Question:** Why does energy decrease at each trophic level?
- **Key Answer:** Only about 10% of energy is transferred from one trophic level to the next. The rest is lost as heat through cellular respiration. This limits food chains to 4–5 levels.